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Studierichting: Informatica
Oefeningenreeks 3G nr. 15.4

$$f(x) = \sin x + \frac{1}{2} \sin 2x$$

$$\text{dom } f = \mathbb{R}$$

$$\text{Nulpunten: } \sin x + \frac{1}{2} \sin 2x = 0$$

$$\Leftrightarrow \sin x + \frac{1}{2}(2 \sin x \cos x) = 0$$

$$\Leftrightarrow \sin x + \sin x \cos x = 0$$

$$\Leftrightarrow \sin x(1 + \cos x) = 0$$

$$\Rightarrow \sin x = 0 \vee \cos x = -1$$

$$\Rightarrow x = n\pi \wedge x = \pi + 2k\pi$$

$$\Rightarrow x \in \mathbb{R}, n \in \mathbb{Z} | x = n\pi$$

$$\text{Periode: } 2\pi$$

VA:

geen VA.

HA:

Geen HA.

SA:

geen SA.

Tekenonderzoek:

$$f'(x) = \frac{d}{dx} \left(\sin x + \frac{1}{2} \sin 2x \right)$$

$$= \cos x + \frac{1}{2} \cdot 2 \cos 2x$$

$$= \cos x + \cos 2x$$

$$= 2 \cos \left(\frac{3x}{2} \right) \cos \left(\frac{x}{2} \right)$$

$$2 \cos \left(\frac{3x}{2} \right) \cos \left(\frac{x}{2} \right) = 0 \Leftrightarrow \cos \left(\frac{3x}{2} \right) = 0 \vee \cos \left(\frac{x}{2} \right) = 0$$

$$\Rightarrow x = \frac{\pi}{3} + \frac{2k\pi}{3} \wedge x = \pi + 2k\pi$$

$$\Rightarrow x \in \mathbb{R}, n \in \mathbb{Z} \mid \frac{\pi}{3} + \frac{2n\pi}{3}$$

$$f''(x) = \frac{d}{dx}(\cos x + \cos 2x)$$

$$= -\sin x - 2 \sin 2x$$

$$= -\sin x - 4 \sin x \cos x$$

$$= \sin x(-1 - 4 \cos x)$$

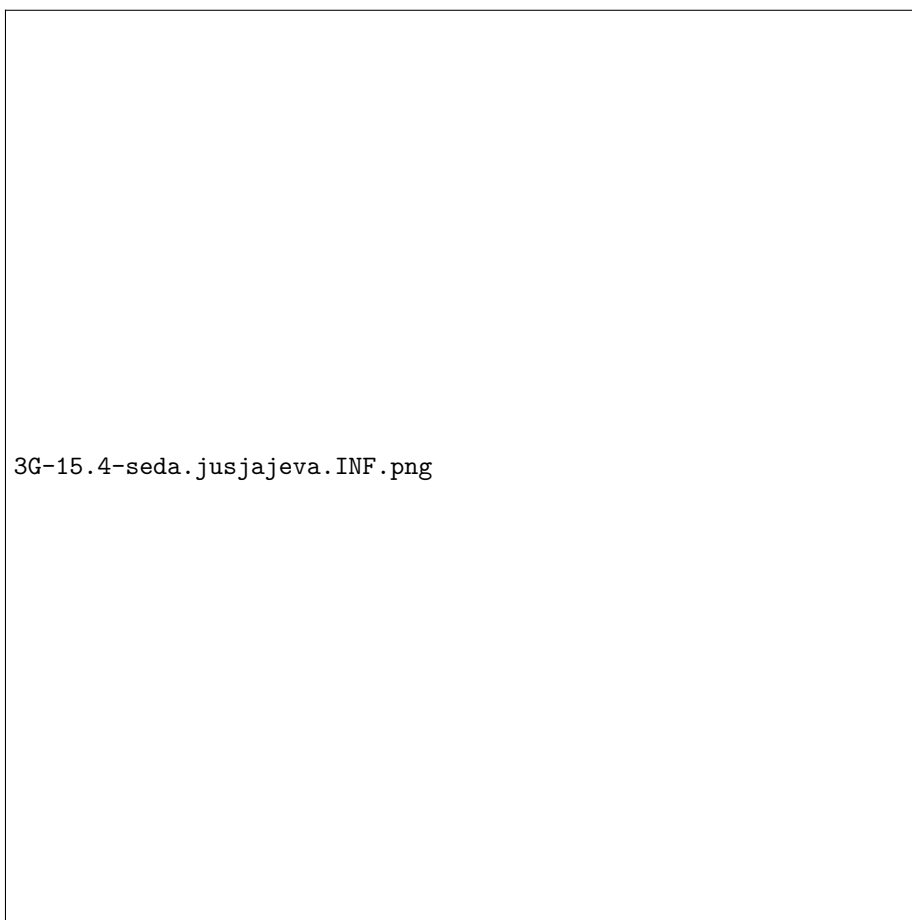
$$\sin x(-1 - 4 \cos x) = 0 \Leftrightarrow \sin x = 0 \vee \cos x = -\frac{1}{4}$$

$$\Rightarrow x = k\pi \wedge x = \pm \text{Bgc} \cos \left(-\frac{1}{4} \right) + 2k\pi$$

x	0			$\frac{\pi}{3}$	$\text{Bgc} \cos \left(-\frac{1}{4} \right)$			π	$2\pi - \text{Bgc} \cos \left(-\frac{1}{4} \right)$		
$f(x)$	−	0	+	+	+	+	−	0	−	−	−
$f'(x)$	+	+	+	0	−	−	−	0	−	−	−
$f''(x)$	+	0	−	−	−	0	+	0	−	0	+
$f(x)$	$\nearrow$	$\nearrow$	$\nearrow$	$M(\sqrt{3})$	$\searrow$	$\searrow$	$\searrow$	$\searrow$	$\searrow$	$\searrow$	$\nearrow$
	$\smile$	$B(0)$	$\frown$	$\frown$		$B\left(\frac{3\sqrt{15}}{8}\right)$	$\smile$	$B(0)$	$\frown$	$\smile$	$B\left(-\frac{3\sqrt{15}}{16}\right)$

Grafiek:

References



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